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ON CORRELATION FUNCTIONS OF THE XY MODEL

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Dedicated to the memory of V. N. Popov

New determinant representations are obtained for the simplest correlation functions of the Heisenberg XY spin chain. Bibliography: 8 titles.

INTRODUCTION

We calculate the simplest equal-time correlation functions at zero temperature for the anisotropic Heisenberg XY chain. This model was first introduced and investigated in [1]. Since then, it has been studied by many authors (see, e.g., [2, 3] and the references given therein). The peculiarity of our approach is that it gives an expression for the correlators in terms of determinants of matrices of order equal to the total number of chain sites (in the case of a finite chain). In the thermodynamic limit, the correlators are represented as the Fredholm determinants of the corresponding linear operators. In [4], representations of this type were first obtained in the case of the gas of impenetrable bosons. The systematic application of this approach to the special case of the isotropic XXO chain [5–7] resulted in deriving integrable differential equations for the temperature time-dependent correlation functions and in explicit calculation of their asymptotics at large time and distances. In the future, we expect to realize this program also in the case of an anisotropic chain.

EIGENFUNCTIONS OF THE HAMILTONIAN

The Hamiltonian of the XY chain describing the integration of the nearest neighbors (of spin 1/2) located at the sites of the one-dimensional periodic lattice can be written in the form

$$H = H_0 + \gamma H_1 - hS_z, \tag{1.1}$$

where

$$H_0 = -\frac{1}{2}\sum_{m=1}^{M}\left(\sigma_+^{(m)}\sigma_-^{(m+1)} + \sigma_-^{(m)}\sigma_+^{(m+1)}\right), \tag{1.2}$$

$$H_1 = -\frac{1}{2}\sum_{m=1}^{M}\left(\sigma_+^{(m)}\sigma_+^{(m+1)} + \sigma_-^{(m)}\sigma_-^{(m+1)}\right), \tag{1.3}$$

$$S_z = \frac{1}{2}\sum_{m=1}^{M}\sigma_z^{(m)}. \tag{1.4}$$

Here $\sigma_\pm = \frac{1}{2}\left(\sigma_x \pm i\sigma_y\right)$ and σ_z are the Pauli matrices with the commutation relations

$$\left[\sigma_a^{(m)}, \sigma_b^{(n)}\right] = 2i\delta_{mn}\in_{abc}\sigma_c^{(m)}, \quad \sigma_a^{(M+1)} \equiv \sigma_a^{(1)},$$

γ is the anisotropy parameter in the XY plane, h is the constant external magnetic field, and S_z is the operator of the third component of the total spin. The total number M of lattice sites is chosen to be even.

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The case $\gamma = 0$ corresponds to the XXO model. The eigenfunctions of its Hamiltonian are well known as a special simple form (with the two-particle scattering phases equal to zero) of the Bethe ansatz for the XXZ chain. We write them in the form given in [5]. The eigenvectors $|\Phi_N\rangle$,

$$(H_0 - \gamma S_z) \mid \Phi_N(\{p\})\rangle = E_N^{(0)}(\{p\}) \mid \Phi_N(\{p\})\rangle, \quad E_N(\{p\}) = \sum_{a=1}^{N} \varepsilon_0(p_a), \quad \varepsilon_0(p) = h - \cos p, \tag{1.5}$$

are parametrized by sets of N different quasimomenta p_a, $-\pi \leq p_a \leq \pi$. Their permitted values (obtained from the periodicity condition for the wave functions) depend on the parity of the number N of quasiparticles,

$$p_a = \frac{2\pi}{M}\left(-\frac{M}{2} + j_a - \frac{1}{4}\left(1 + (-1)^N\right)\right), \tag{1.6}$$

where j_a are integers, $1 \leq j_a \leq M$. We have

$$\left|\Phi_N(\{p\})\right\rangle = \frac{1}{\sqrt{N!M^N}} \sum_{m_1,\ldots,m_N=1}^{M} \chi_N(\{p\}|\{m\})\sigma_-^{(m_1)} \ldots \sigma^{(m_N)}|0\rangle, \tag{1.7}$$

with

$$\chi_N(\{p\} \mid \{m\}) = \frac{1}{\sqrt{N!}}\left(\prod_{1\leq a<b\leq N} \in (m_b - m_a)\right) \det_N\left(\exp\{im_a p_b\}\right), \tag{1.8}$$

where $|0\rangle \equiv \bigotimes_{m=1}^{M} |\uparrow_m\rangle$ is the ferromagnetic state (all the spins up),

$$(H_0 - hS_z) \mid 0\rangle = -\frac{Mh}{2} \mid 0\rangle,$$

and $a \in (m)$ denotes the sign function ($\in (m) = \operatorname{sgn}(m)$). Distinct eigenfunctions correspond to distinct (up to permutations) sets of quasimomenta. The total number of the eigenfunctions is equal to 2^M. Eigenfunctions corresponding to distinct (up to permutations) sets of momenta are orthogonal and the following normalization condition holds:

$$\langle \Phi_N(\{p\}) \mid \Phi_N(\{p\})\rangle = 1. \tag{1.9}$$

We proceed to construct the eigenfunctions of the XY chain, using the basis consisting of the eigenfunctions of the XXO chain. To do this, it is necessary to calculate the matrix elements of the operator H_1 (1.3) in this basis. It can easily be seen that, e.g., for the operator $\sigma_-^{(m)}\sigma_-^{(m+1)}$, only the following matrix elements are not equal to zero (up to permutations of momenta):

$$\begin{aligned}
&\langle \Phi_{N+2}(p_1, \ldots, p_N, q_{N+1}, q_{N+2})\left|\sigma_-^{(m)}\sigma_-^{(m+1)}\right|\Phi_N(p_1, \ldots, p_N)\rangle = \\
&= \frac{1}{M}\exp\left\{-im(q_{N+2} + q_{N+1})\right\}\left(\exp\left\{-iq_{N+2}\right\} - \exp\left\{-iq_{N+1}\right\}\right);
\end{aligned}$$

a similar expression is also valid for the matrix elements of the operator $\sigma_+^{(m)}\sigma_+^{(m+1)}$. Summing over m and using relation (1.6), we obtain the following relations for the nonzero matrix elements of the operator H_1:

$$\begin{aligned}
&\langle \Phi_{N+2}(p_1, \ldots, p_N, p, -p)\left|H_1\right|\Phi_N(p_1, \ldots, p_N)\rangle = -i\sin p \\
&\langle \Phi_{N-2}(p_1, \ldots, p_{N-2})\left|H_1\right|\Phi_N(p_1, \ldots, p_{N-2}, p, -p)\rangle = i\sin p.
\end{aligned} \tag{1.10}$$

Hence the 2^M-dimensional space $\mathcal{H}$, where the Hamiltonian H (1.1) acts, can be decomposed into a direct sum of subspaces invariant under this action. These subspaces are linear spans of the basis state vectors

$$\left|\Phi_{N+2n}(p_1, \ldots, p_N; q_{i_1}, -q_{i_1}, \ldots, q_{i_n}, -q_{i_n})\right\rangle$$

of the XXO chain. Here, the momenta $p_1, \dots, p_N$, $|p_a| \neq |p_b|$ $(a \neq b)$ (fixed for a given subspace up to a permutation) and the momenta $q_{i_m} > 0$, $q_{i_m} > q_{i_l}$ $(m > l)$, $q_{i_m} > q_{k_m}$ $(i_m > k_m)$, $q_{i_m} \neq |p_a|$ take values (1.6) permitted by the periodicity conditions. The dimension of the subspace is 2^K, where K is the largest possible number of pairs $(q, -q)$ of quasiparticles with $q \neq |p_a|$ (e.g., $K = (M/2) - N$ if N is even).

This observation allows us to construct the complete orthogonal set of eigenfunctions $\mid \Psi_N\rangle$,

$$H \mid \Psi_N\rangle = E_N \mid \Psi_N\rangle, \tag{1.11}$$

of the Hamiltonian H parametrized by the momenta $p_1, \dots, p_N$ and by the variables $r_1, \dots, r_K$ taking two values $(r_i = \pm 1)$. These eigenfunctions have the form

$$\begin{aligned} \mid \Psi_N(p_1, \dots, p_N; r_1, \dots, r_K)\rangle = \mid \Psi_N(p_1, \dots, p_N)\rangle + \sum_{n=1}^{K} \frac{1}{n!} \sum_{i_1, \dots, i_n = 1}^{K} \Bigl(\prod_{l=1}^{n} \alpha(q_{i_l}, r_{i_l}) \Bigr) \times \\ \times \mid \Phi_{N+2n}(p_1, \dots, p_N; q_{i_1}, -q_{i_1}, \dots, q_{i_n}, -q_{i_n})\rangle \end{aligned} \tag{1.12}$$

(it should be noted that the state vectors $\mid \Phi_{N+2n}\rangle$ are symmetric under the permutations $(q_{i_l}, -q_{i_l}) \leftrightarrow (q_{i_m}, -q_{i_m})$ and are equal to zero if at least two momenta coincide). The function α in (1.12) is given by the formula

$$\alpha(q, r) = \frac{1}{i\gamma \sin q} \bigl(\varepsilon_0(q) + r\varepsilon(q) \bigr), \tag{1.13}$$

where $\varepsilon_0(q) = h - \cos q$ (see (1.5)) and

$$\varepsilon(q) = \sqrt{\varepsilon_0^2(q) + \gamma^2 \sin^2 q} \geq 0. \tag{1.14}$$

The orthogonality of the state vectors $\mid \Psi\rangle$ with different sets of momenta $\{p\}$ and $\{p'\}$ is obvious since the corresponding basis vectors $\mid \Phi\rangle$ are orthogonal. If $\{p\} = \{p'\}$ $(\{p\} \equiv p_1, \dots, p_N)$, then, taking into account (1.9), we easily obtain

$$\langle \Psi_N(\{p\}, \{r\}) \mid \Psi_N(\{p\}, \{\widetilde{r}\})\rangle = \prod_{i=1}^{K} \bigl(1 + \alpha^*(q_i, r_i)\alpha(q_i, \widetilde{r}_i) \bigr).$$

Since $\alpha^*(q, r)\alpha(q, -r) = -1$ (see (1.13)), the left-hand side vanishes if at least two of the variables r, $\widetilde{r}$ are distinct. Thus, the state vectors (1.12) form an orthogonal basis of eigenfunctions of the XY chain normalized by the condition

$$\langle \Psi_N(\{p\}, \{r\}) \mid \Psi_N(\{p\}, \{r\})\rangle = \prod_{i=1}^{K} \bigl(1 + |\alpha(q_i, r_i)|^2 \bigr). \tag{1.15}$$

For the eigenvalues E_N of the Hamiltonian on the periodic lattice, we obtain the expression

$$E_N = \sum_{i=1}^{K} \varepsilon_0(q_i) + \sum_{a=1}^{N} \varepsilon_0(p_a) + \sum_{i=1}^{K} r_i \varepsilon_i - \frac{Mh}{2}. \tag{1.16}$$

Due to (1.6), the first two sums in (1.16) can be calculated in an explicit form. If N is even, we obtain

$$E_N = \sum_{i=1}^{K} r_i \varepsilon(q_i). \tag{1.17}$$

If N is odd, the answer depends on whether the momenta 0 and π are among the momenta p (such momenta cannot be among the momenta q):

$$E_N = \sum_{i=1}^{K} r_i\varepsilon(q_i) - h, \ N\text{—odd}, \quad 0,\pi \notin \{p\} \tag{1.18}$$

$$E_N = \sum_{i=1}^{K} r_i\varepsilon(q_i) + h, \ N\text{—odd}, \quad 0,\pi \in \{p\} \tag{1.19}$$

$$E_N = \sum_{i=1}^{K} r_i\varepsilon(q_i) - 1, \ N\text{—odd}, \ 0 \in \{p\}, \ \pi \notin \{p\} \tag{1.20}$$

$$E_N = \sum_{i=1}^{K} r_i\varepsilon(q_i) + 1, \ N\text{—odd}, \ 0 \notin \{p\}, \ \pi \in \{p\}. \tag{1.21}$$

Let us explain briefly how to obtain a formula for the eigenfunctions. Substituting expression (1.12) into Eq. (1.11), we see that it "splits" into K equations for α,

$$-i\gamma\alpha^2(q_j,r_j)\sin q_j + 2\varepsilon_0(q_j)\alpha(q_j,r_j) - i\gamma\sin q_j = 0, \ j = 1,\dots,K,$$

and an expression for the energy E_N,

$$E_N = -\frac{Mh}{2} + \sum_{a=1}^{N}\varepsilon_0(p_a) + \sum_{j=1}^{K} i\gamma\alpha(q_j,r_j)\sin q_j.$$

The quantities α (1.13) are solutions of the corresponding quadratic equations, and the expression for the energy leads to (1.17)–(1.21).

Using the basis constructed above, we can easily calculate the partition function Z,

$$Z_M = \mathrm{Tr}\left(\exp\{-\frac{H}{T}\}\right) = \sum_{N}\sum_{p_1,\dots,p_N}\sum_{r_1,\dots,r_K=\pm 1}\exp\{-E_N/T\} \tag{1.22}$$

for the XY chain with even number M of sites. The energy values E_N (1.17) are 2^N times degenerate with respect to the changing of signs of the momenta $p_1,\dots,p_N$ unequal to zero or π. Therefore, putting $r_a \equiv r(p_a) \equiv 0$ $(a = 1,\dots,N)$, we can extend the summation in formulas (1.18)–(1.21) over all nonnegative momenta permitted by Eq. (1.6). For example, in the case (1.18), we have 2^N eigenfunctions with the given set $r_1,\dots,r_{M/2}$ $(r = 0,\pm 1)$, $N = -\sum_{i=1}^{M/2}|r_i| + \frac{M}{2}$, and the following expression for the corresponding contribution to the partition function:

$$\sum_{r_1,\dots,r_{M/2}=0,\pm 1} 2^{N-1}\exp\left(-\frac{1}{T}\sum_{i=1}^{M/2} r_i\varepsilon_i\right)\left(1+(-1)^{\sum_{i=1}^{M/2}(1-|r_i|)}\right) = 2^{\frac{M}{2}-1}\left(\prod_{i=1}^{M/2}\left(1+\cosh\frac{\varepsilon_i}{T}\right)+\prod_{i=1}^{M/2}\left(\cosh\frac{\varepsilon_i}{T}-1\right)\right).$$

Here $\varepsilon_i = \varepsilon(p_i)$ and $p_i = \frac{2\pi}{M}\left(-\frac{1}{2}+i\right)$, (see (1.6), (1.14)). Similarly, we calculate the contributions of the cases (1.19)–(1.21). Finally, we obtain

$$\begin{aligned} Z_M = 2^{\frac{M}{2}-1}\Bigg\{&\left[\prod_{i=1}^{M/2}\left(\cosh\frac{\varepsilon_i}{T}+1\right)+\prod_{i=1}^{M/2}\cosh\left(\left(\frac{\varepsilon_i}{T}-1\right)\right]+\right. \\ &+\cosh\frac{h}{T}\left[\prod_{i=1}^{M/2}\left(\cosh\frac{\widetilde{\varepsilon}_i}{T}+1\right)-\prod_{i=1}^{M/2-1}\left(\cosh\frac{\widetilde{\varepsilon}_i}{T}-1\right)\right]+ \\ &\left.+\cosh\frac{1}{T}\left[\prod_{i=1}^{M/2-1}\left(\cosh\frac{\widetilde{\varepsilon}_i}{T}+1\right)+\prod_{i=1}^{M/2-1}\left(\cosh\frac{\widetilde{\varepsilon}_i}{T}-1\right)\right]\right\}. \end{aligned} \tag{1.23}$$

Here $\widetilde{\varepsilon}_j = \varepsilon(\widetilde{p}_j)$ and $\widetilde{p}_j = \frac{2\pi}{M} j$.

In the thermodynamic limit ($M \to \infty$), we obtain the well-known expressions [1] for the free energy on the infinite lattice,

$$F(T,h) = -\lim_{M\to\infty} \frac{T}{M} \ln Z = -\frac{T}{2\pi} \int_0^\pi \ln 2\big(\cosh \frac{\varepsilon(p)}{T} + 1\big)\, dp,$$

and for magnetization along the third axis,

$$M_z = -\frac{\partial}{\partial h} F(T,h) = \frac{1}{2\pi} \int_0^\pi \tanh\big(\frac{\varepsilon_0(p)}{2T}\big) \frac{\varepsilon_0(p)}{\varepsilon(p)}\, dp. \tag{1.24}$$

Generating functional of the correlators of the third spin components

In this section, we calculate the simplest correlation function in the case of zero temperature, namely, the mean value of the operator $\exp\{\alpha Q(m)\}$ normalized with respect to the ground state $|\Psi_0\rangle$ of the XY model. The ground state of the XY model is constructed by filling the ferromagnetic state by the pairs of quasiparticles with $r_i = -1$ (see (1.6), (1.12), (1.17))

$$|\Psi_0\rangle = |0\rangle + \sum_{n=1}^{M/2} \frac{1}{n!} \sum_{i_1,\dots,i_n=1}^{K} \Big(\prod_{l=1}^{n} \alpha(q_{i_l}, -1)\Big) \times |\Phi_{2n}(q_{i_1}, -q_{i_1}, \dots, q_{i_n}, -q_{i_n})\rangle. \tag{2.1}$$

The operator $Q(m)$ of the number of quasiparticles at the first m sites is given by the formula (see, e.g., [5])

$$Q(m) = \frac{1}{2} \sum_{j=1}^{m} \big(1 - \sigma_z^{(j)}\big) = \sum_{j=1}^{m} \sigma_-^{(j)} \sigma_+^{(j)}. \tag{2.2}$$

The form factors of the operator $\exp\{\alpha Q(m)\}$ in the basis consisting of eigenfunctions of the XXO model can easily be calculated much as the diagonal matrix elements were calculated in [5]. Nonzero matrix elements have the following form:

$$\langle \Phi_N(p_1, \dots, p_N) \mid \exp\{\alpha Q(m)\} \mid \Phi_N(p'_1, \dots, p'_N)\rangle = \det_N \mathcal{M}(\{p\}, \{p'\}). \tag{2.3}$$

Here $\mathcal{M}$ is the $N \times N$ matrix with the matrix elements

$$\mathcal{M}_{ij} = \delta_{p_i p'_j} + \frac{\beta}{M} F(p_i, p'_j); \quad F(p,q) \equiv \frac{\sin\frac{m}{2}(p-q)}{\sin\frac{1}{2}(p-q)}; \quad \beta \equiv e^\alpha - 1. \tag{2.4}$$

Due to these formulas and normalizations (1.9) and (1.15), the mean value with respect to the ground state of the XY model on the lattice with M sites can be represented in the following form:

$$\langle \exp\{\alpha Q(m)\}\rangle_M \equiv \frac{\langle \Psi_0 | \exp\{\alpha Q(m)\} | \Psi_0\rangle}{\langle \Psi_0 | \Psi_0\rangle} =$$

$$= \Big(\prod_{p>0} (1 + |\alpha(p)|)^{-1}\Big) \sum_{N=0}^{M/2} \big(\frac{1}{N!}\big)^2 \times \sum_{\substack{\{p_i\}_N, \{q_j\}_N \\ p_i>0, q_j>0}} \Big(\prod_{i=1}^{N} |\alpha(p_i)|\Big)\Big(\prod_{j=1}^{N} |\alpha(q_i)|\Big) \times \tag{2.5}$$

$$\times \det_{2N} \mathcal{M}(\{p,-p\}, \{q,-q\}); \quad \alpha(p) \equiv \alpha(p,-1).$$

Here the sum is taken over all ordered sets of different positive values of quasimomentum permitted by (1.6).

Due to the symmetry of the function F (2.4), $F(p,q) = F(-p,-q)$, we can represent the $2N \times 2N$ matrix $\mathcal{M}$ in the block form

$$\mathcal{M} = \begin{pmatrix} B & C \\ C & B \end{pmatrix}, \tag{2.6}$$

where B and C are the $N \times N$ matrices with the matrix elements

$$B_{ij} = \delta_{p_i q_j} + \frac{\beta}{M} F(p_i, q_j); \quad C_{ij} = \frac{\beta}{M} F(p_i, -q_j). \tag{2.7}$$

The determinant of $\mathcal{M}$ can be rewritten as a product of two determinants of $N \times N$ matrices,

$$\det_{2N} \mathcal{M} = \det_N (B + C) \det_N (B - C). \tag{2.8}$$

Using this representation and the fact that the determinant is linear with respect to the matrix elements of any row or column, one can easily calculate the sum over the momenta p in (2.5).

We have

$$\frac{1}{N!} \sum_{\{p_i\}_N, p_i > 0} \left(\prod_{i=1}^{N} |\alpha(p_i)| \right) \det_{2N} \mathcal{M}(\{p,-p\},\{q,-q\}) = \det_N G, \tag{2.9}$$

where the $N \times N$ matrix G has the form

$$G_{jk} = \sum_{p>0} |\alpha(p)| \left(\delta_{p,q_j} + \frac{\beta}{M} (F(p,q_j) + F(p,-q_j)) \right) \times \left(\delta_{p,q_k} + \frac{\beta}{M} (F(p,q_k) + F(p,-q_k)) \right). \tag{2.10}$$

Using simple transformations, we can represent the correlator as the determinant of the $(M/2) \times (M/2)$ matrix,

$$\langle \exp\{\alpha Q(m)\} \rangle_M = \prod_{i=1}^{M/2} (1 + |\alpha(p_i))^{-1} \times \det_{M/2} (I + F_+ A F_- A), \tag{2.11}$$

where the diagonal $(M/2) \times (M/2)$ matrix A and the matrices F_+ and F_- are defined by the formulas

$$A = \mathrm{diag}(|\alpha(p_i)|); \quad (F_\pm)_{jk} = \delta_{jk} + \frac{\beta}{M} (F(q_j, q_k) \pm F(q_j, -q_k)). \tag{2.12}$$

Using the well-known formula

$$\det_M \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det_{M/2} (AD - BD^{-1}CD), \tag{2.13}$$

we can represent this result in a more convenient form of $M \times M$ matrices. We have

$$\det_{M/2} (I + F_+ A F_- A) = \det_M \begin{pmatrix} I, & F_+ A \\ F_- A, & I \end{pmatrix}.$$

Performing the similarity transformation with the matrix

$$e^{-\frac{i\pi}{4}\sigma_1} \otimes I_{M/2} = \frac{1}{\sqrt{2}} \begin{pmatrix} I, & iI \\ iI, & I \end{pmatrix},$$

we can easily obtain the final formula for the correlator on the lattice with a finite number of sites,

$$\langle \exp\{\alpha Q(m)\} \rangle_M = \det_M (I + \widetilde{\mathcal{M}}). \tag{2.14}$$

Here I is the identity $M \times M$ matrix and the $M \times M$ matrix $\widetilde{\mathcal{M}}$ has the following form ($\beta \equiv (e^{\alpha}-1)/M$):

$$\widetilde{\mathcal{M}}(p,q) = \frac{\beta}{M} F(p,q) \frac{\alpha(q)}{\alpha(q)+1} \equiv \frac{\beta}{M} F(p,q)(1+e^{i\varphi(q)}) \tag{2.15}$$

(we have taken into account that, in the ground state, $\alpha(q) = i|\alpha(q)|$, $q > 0$ and $\alpha(q) = -i|\alpha(q)|$, $q < 0$). Here we also use the following convenient notation:

$$\varphi(p) = \frac{1}{i} \ln \frac{\cos p - h + i\gamma \sin p}{\sqrt{(\cos p - h)^2 + \gamma^2 \sin^2 p}},$$
$$\alpha(p, \pm 1) \equiv (-i \tan(\varphi(p)/2))^{\pm 1}; \quad \alpha(p) \equiv \alpha(p,-1). \tag{2.16}$$

It should be noted that the only distinction between (2.15) and the corresponding formula for the XXO case is in the additional "weight" $(1+e^{i\varphi(q)})$, and this distinction disappears for $\gamma \to 0$.

Now we discuss the thermodynamic limit ($M \to \infty$, the magnetic field h is fixed). In this limit, one should replace the right-hand side of formula (2.14) by the Fredholm determinant of the linear integral operator $\widehat{\mathcal{M}}$ acting on functions on the interval $[-\pi, \pi]$,

$$(\widehat{\mathcal{M}} f)(p) = \int_{-\pi}^{\pi} \widetilde{\mathcal{M}}(p,q) f(q) dq \tag{2.17}$$

(this is explained in detail in [5]). The kernel of this integral operator is given by (2.15), and in the thermodynamic limit, the correlator can be represented as the Fredholm determinant

$$\langle \exp\{\alpha Q(m)\} \rangle = \det(\widehat{I} + \widehat{\mathcal{M}}), \tag{2.18}$$

where $\widehat{I}$ is the identity operator on the interval $[-\pi, \pi]$.

The correlator allows us to calculate local correlators of the third spin components. In particular, the magnetization $\langle \sigma_z^{(m)} \rangle$ can be represented as follows:

$$\langle \sigma_z^{(m)} \rangle_M = -2 D_1 \frac{d}{\alpha \alpha} \langle \exp\{\alpha Q(m)\} \rangle_M \big|_{\alpha=0} + 1,$$

where D_1 is the operator of the first derivative on the lattice, $(D_1 f)(m) \equiv f(m+1) - f(m)$. Thus, we obtain

$$\langle \sigma_z^{(m)} \rangle_M = 1 - \frac{4}{M} \sum_{p>0} \cos^2 \frac{\varphi(p)}{2} \underset{M \to \infty}{\to} -\frac{1}{\pi} \int_0^{\pi} \cos \varphi(p) dp, \tag{2.19}$$

which coincides with (1.24) for $T = 0$.

For the correlator of the third spin components, we have

$$\langle \sigma_z^{(m+1)} \sigma_z^{(m)} \rangle_M = 4 D_2 \frac{d^2}{d\alpha^2} \langle \exp\{\alpha Q(m)\} \rangle \big|_{\alpha=0} + 2 \langle \sigma_z^{(m)} \rangle_M - 1,$$

where $(D_2 f)(m) \equiv f(m+1) + f(m-1) - 2f(m)$ is the operator of the second derivative on the lattice. After some calculation, we obtain the following result for the finite lattice:

$$\langle \sigma_z^{(m+1)} \sigma_z^{(m)} \rangle_M = \langle \sigma_z^{(m)} \rangle_M^2 + \frac{1}{M^2} \sum_{-\pi < p,q < \pi} \cos m(p-q) [\sin \varphi(p) \sin \varphi(q) - (1+\cos \varphi(p))(1+\cos \varphi(q))],$$

which, in the thermodynamic limit, leads to the well-known formula [8]:

$$\langle \sigma_z^{(m+1)} \sigma_z^{(1)} \rangle = \langle \sigma_z \rangle^2 - \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} dp \cos \varphi(p) e^{imp} \right)^2 - \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} dp \sin \varphi(p) e^{imp} \right)^2$$

CORRELATOR $\langle\sigma_+^{m+1}\sigma_-^{(1)}\rangle$

In this section, we use the same method to calculate the equal-time correlator of local matrices σ_+ and σ_-,

$$\langle\sigma_+^{(m+1)}\sigma_-^{(1)}\rangle_M = \frac{\langle\Psi_0|\sigma_+^{(m+1)}\sigma_-^{(1)}|\Psi_0\rangle}{\langle\Psi_0|\Psi_0\rangle}. \tag{3.1}$$

By a simple generalization of the results of [5], one can calculate the corresponding form factors in the basis of eigenfunction of the XXO model,

$$\langle\Phi_N(p_1,\dots,p_N)|\sigma_+^{(m+1)}\sigma_-^{(1)}|\Phi_N(p'_1,\dots,p'_N)\rangle = \frac{\partial}{\partial x}\det{}_N(S+xR)\big|_{x=0}. \tag{3.2}$$

Here $S(\{p\},\{p'\})$ and $R(\{p\},\{p'\})$ are the $N\times N$ matrices with the entries

$$S_{ij} = \delta_{p_i,p'_j} - \frac{2}{M}\frac{\sin\frac{m}{2}(p_i-p'_j)}{\tan\frac{1}{2}(p_j-p'_j)}, \tag{3.3}$$

$$R_{ij} = \frac{1}{M}\exp\left\{\frac{im}{2}(p_j+p'_j)\right\}, \tag{3.4}$$

satisfying the relations

$$S(p,q) = S(-p,-q), \quad R^*(p,q) = R(-p,-q), \tag{3.5}$$

where the asterisk stands for the complex conjugation. One can easily obtain the following expression for the correlator in question:

$$\langle\sigma_+^{(m+1)}\sigma_-^{(1)}\rangle_M = \prod_{k=1}^{M/2}(1+|\alpha(p_k)|^2)^{-1}\times\sum_{N=0}^{M/2}\left(\frac{1}{N!}\right)^2\sum_{\substack{\{p_i\}_N,\{q_j\}_N\\ p_i>0,q_j>0}}\left(\prod_{i=1}^{N}|\alpha(p_i)|\right)\left(\prod_{j=1}^{N}|\alpha(q_j)|\right)\times$$

$$\times\frac{\partial}{\partial x}\det{}_{2N}(S(\{p,-p\},\{q,-q\})+xR(\{p,-p\},\{q,-q\}))\big|_{x=0}. \tag{3.6}$$

Here, the summation is performed as in (2.8). Taking into account (3.5), we see that the matrix on the right-hand side has the form

$$S+xR = \begin{pmatrix} A+xC, & B+xD \\ B+xD^*, & A+xC^* \end{pmatrix}, \tag{3.7}$$

where the entries of $N\times N$ matrices A, B, C, and D are equal to

$$A_{ij} = \delta_{p_iq_j} - \frac{2}{M}\frac{\sin\frac{m}{2}(p_i-q_j)}{\tan\frac{1}{2}(p_i-q_j)}, \quad C_{ij} = \frac{1}{M}\exp\left\{\frac{im}{2}(p_i+q_j)\right\},$$

$$B_{ij} = -\frac{2}{M}\frac{\sin\frac{m}{2}(p_i+q_j)}{\tan\frac{1}{2}(p_i+q_j)}, \qquad D_{ij} = \exp\left\{\frac{im}{2}(p_i-q_j)\right\}.$$

It can easily be verified that

$$\frac{\partial}{\partial x}\det\begin{pmatrix} A+xC, & B+xD \\ B+xD^*, & A+xC^* \end{pmatrix}\Big|_{x=0} = \frac{\partial}{\partial x}\det\begin{pmatrix} A+\frac{x}{2}(C+C^*), & B+\frac{x}{2}(D+D^*) \\ B+\frac{x}{2}(D+D^*), & A+\frac{x}{2}(C+C^*) \end{pmatrix}\Big|_{x=0}$$

(it should be noted that the matrix on left-hand side depends linearly on the parameter x). Therefore, in (3.6) , we have

$$\frac{\partial}{\partial x}\det(S+xR)\big|_{x=0} = \frac{\partial}{\partial x}\det\mathcal{M}_1(\{p,-p\},\{q,-q\})\big|_{x=0},$$

where

$$\mathcal{M}_1(p,q) = \delta_{p,q} + \frac{1}{M} F_1(p,q); \quad F_1(p,q) = -\frac{2\sin\frac{m}{2}(p-q)}{\tan\frac{1}{2}(p-q)} + x\cos\frac{m}{2}(p-q), \tag{3.8}$$

Note that, calculating the correlator $\langle\exp\{\alpha Q(m)\}\rangle$ in the previous section, we did not use the explicit form of $F(p,q)$ (2.4) but only its symmetry, $F(p,q) = F(-p,-q)$. The function $F_1(p,q)$ has the same property. Therefore, in the same way we obtain the result for the correlator on the finite lattice,

$$\langle\sigma_+^{(m+1)}\sigma_-^{(1)}\rangle_M = \frac{\partial}{\partial x}\det_M\left(U + xP\right)\Big|_{x=0}. \tag{3.9}$$

Here the entries of $M\times M$ matrices U and P are equal to

$$U(p,q) = \delta_{p,q} - \frac{1}{M}\frac{\sin\frac{m}{2}(p-q)}{\tan\frac{1}{2}(p-q)}\left(\exp\{i\varphi(q)\}+1\right), \quad P(p,q) = \frac{1}{2M}\cos\frac{m}{2}(p+q)\left(\exp\{i\varphi(q)\}+1\right). \tag{3.10}$$

The matrix elements of these matrices are numbered by the values of the momenta p and q permitted by (1.6) with $N = M$ (M is even); the value $\varphi(q)$ is defined in (2.16).

In the thermodynamic limit ($M \to \infty$), the correlator can be expressed as the Fredholm determinant

$$\langle\sigma_+^{(m+1)}\sigma_-^{(1)}\rangle_M = \frac{\partial}{\partial x}\det\left(I + \widehat{V} + x\widehat{P}\right)\Big|_{x=0} \tag{3.11}$$

of the following linear integral operators acting on functions on the interval $[-\pi,\pi]$:

$$(\widehat{V}f)(p) = \int_{-\pi}^{\pi} V(p,q)f(q)dq, \quad (\widehat{P}f)(p) = \int_{-\pi}^{\pi} P(p,q)f(q)dq$$

with kernels

$$V(p,q) = -\frac{1}{2\pi}\frac{\sin\frac{m}{2}(p-q)}{\tan\frac{1}{2}(p-q)}\left(\exp\{i\varphi(q)\}+1\right) \quad \text{and} \quad P(p,q) = \frac{1}{4\pi}\cos(p,q)\left(\exp\{i\varphi(q)\}+1\right).$$

In the limit $\gamma = 0$, all formulas transform into the corresponding expressions for the XXO chain obtained in [5].

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